

## Professional Summary

Final-year B.Tech (Data Science) student and Software Development Intern building full-stack, AI-powered BFSI platforms. Skilled in scalable asynchronous REST APIs, large-scale data pipelines, LLM/RAG applications, and machine-learning systems using FastAPI, React, TypeScript, and PostgreSQL.

## Professional Experience

### Software Development Intern

Feb 2026 – Present

*ATS Services, Faridabad*

- Architected and built a production BFSI debt-collection and insurance-retention analytics platform from scratch — an async FastAPI + PostgreSQL backend and a React 18 + TypeScript (Vite) SPA on a strict controller-service-repository architecture
- Engineered an automated Oracle-to-PostgreSQL sync service: a schema-mapping layer converts raw data from a live Oracle database into the warehouse model, run nightly at 2 AM by a PostgreSQL-native worker (`FOR UPDATE SKIP LOCKED`) that incrementally loads new files and refreshes portal tables for daily-updated analytics
- Processed and reconciled 40 GB+ datasets through staging tables, chunked batch transforms, and table partitioning with index/partition-pruning tuning under `work_mem/statement_timeout` guards
- Built a multi-layer caching strategy — a Postgres-backed analytics cache, in-process TTL caches with generation-based O(1) invalidation, and client-side TanStack Query caching — reducing redundant database load and dashboard latency
- Engineered the AI layer — an LLM-powered (Llama 3) natural-language-to-SQL agent plus ML prediction engines (Random Forest, Logistic Regression) and statistical S-curve forecasting for debt-recovery and policy-lapse propensity
- Secured sensitive financial data with JWT authentication (single-flight refresh), AES-256-CBC payload encryption, CSRF protection, rate limiting, and role-based access control

### IT Project Intern

Jun 2025 – Aug 2025

*AMU Leasing Pvt. Ltd., Noida*

- Built Power BI dashboards for financial analytics, reducing report generation time by 50%
- Integrated CRIF and Aadhaar eKYC APIs for automated customer verification, processing 500+ documents monthly

## Key Projects

### ML & Analytics Portal — BFSI Predictive Platform

*FastAPI, React, Llama 3, PostgreSQL, Random Forest, Recharts*

- Built a full-stack predictive portal with a Glassmorphism UI for debt-collection and life-insurance forecasting, serving real-time Random Forest and Logistic Regression inference
- Implemented a Llama 3 powered SQL Agent for natural-language data querying and interactive Recharts analytics, secured with JWT authentication and Alembic-managed schema migrations

### Local-Hosted Desktop AI Agent

*Python, Tkinter, Ollama (LLM), Whisper, Edge-TTS*

- Built a real-time, offline voice assistant using Whisper (STT), Edge-TTS, and a local Ollama LLM for full data privacy
- Implemented async processing and multithreading for concurrent voice capture and LLM inference, with a Tkinter UI

**Financial Analysis Portal** (*FastAPI, React/Vite, SQL Agent, Recharts*) — Built an analytics dashboard with a natural-language SQL Agent bot and Recharts KPI visualizations over PostgreSQL.

**Airbnb Price Prediction** (*XGBoost, Python, Google Colab*) — Trained an XGBoost regression model for price forecasting with extensive EDA and rigorous feature engineering.

## Technical Skills

**Languages:** Python, SQL, JavaScript, TypeScript

**Backend & APIs:** FastAPI, REST API Design, Asynchronous Programming, SQLAlchemy, Pydantic, Alembic, JWT/OAuth2, bcrypt, Rate Limiting, Caching

**Data & Pipelines:** PostgreSQL, Oracle, MySQL, ETL, Schema Mapping, Data Warehousing, Table Partitioning, SFTP Ingestion, Vector Databases

**Frontend:** React, Vite, TypeScript, Zustand, TanStack Query, Recharts, Tailwind CSS

**AI & ML:** LLM Integration, RAG, LangChain, Llama 3, Ollama, scikit-learn, Random Forest, Logistic Regression, XGBoost

**Tools & DevOps:** Git, Docker, Linux, Postman, Power BI, CI/CD

## Education

### B.Tech in Mathematics and Data Science

2022 – 2026

*NIT - Bhopal*

Class XII: 85% (2022) — Class X: 86% (2020)